Function Machines

Year 4 Pt 2 · **Homework Worksheet**

Worked Example

Function Machine Practice: A function machine doubles a number, then subtracts 4. What is the output when the input is 9?

Answer: 9 × 2 = 18; 18 − 4 = 14.

Function Basics (15 questions)

1. A function machine applies the rule ×7 to its input. What is the output when the input is 8?
2. A function machine applies the rule +12 to its input. What is the output when the input is 25?
3. A function machine applies the rule −9 to its input. What is the output when the input is 30?
4. A function machine applies the rule ÷6 to its input. What is the output when the input is 42? A) 6 B) 7 C) 8 D) 9 E) 10
5. A two-step function machine: input → +5 → ×2 → output. What is the output when the input is 6?
6. A two-step function machine: input → ×3 → +4 → output. What is the output when the input is 8? A) 24 B) 26 C) 28 D) 30 E) 32
7. A function machine applies the rule ×9 to its input. If the output is 54, what was the input?
8. A function machine applies the rule +18 to its input. If the output is 40, what was the input? A) 20 B) 21 C) 22 D) 23 E) 24
9. A function machine applies the rule −11 to its input. If the output is 9, what was the input?
10. A function machine applies the rule ÷7 to its input. If the output is 6, what was the input?
11. Select ALL that apply: a function machine applies the rule ×4 to each input. Which of these input→output pairs are correct? A) 3→12 B) 5→18 C) 6→24 D) 7→28 E) 9→32
12. A two-step function machine: input → ×5 → −6 → output. What is the output when the input is 4?
13. A two-step function machine: input → +9 → ÷3 → output. What is the output when the input is 12?
14. A function machine takes an input, trebles it, then subtracts 5. What is the output when the input is 6? A) 11 B) 12 C) 13 D) 14 E) 15
15. A function machine applies the rule −20 to its input. What is the output when the input is 20?

Applied Functions (15 questions)

1. A function machine: input → ×2 → +11 → output. What is the output when the input is 9?
2. A function machine: input → −7 → ×5 → output. What is the output when the input is 15? A) 35 B) 40 C) 45 D) 50 E) 55
3. A function machine: input → ÷4 → +9 → output. The input is 24. Find the output.
4. A machine takes an input, adds 6, then multiplies by 2. If the output is 28, what was the input? A) 7 B) 8 C) 9 D) 10 E) 11
5. A function machine: input → ×3 → −10 → output. The output is 11. Find the input.

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6. A function machine turns 4 into 15 and 7 into 24 using the same two-step rule "×a, then +b". Find the rule (give both a and b).

7. A function machine: input → ×6 → output. What is the output when the input is 0.5?

8. Jamal puts a number into a machine. The machine doubles the number, then adds 7. The output is 23. What number did Jamal put in? A) 6 B) 7 C) 8 D) 9 E) 10

9. A function machine: input → +10 → output. What is the output when the input is −6?

10. Select ALL that apply: a function machine applies the rule "add 3, then multiply by 3" to inputs. Which input→output pairs are correct? A) 2→15 B) 4→20 C) 5→24 D) 6→28 E) 7→30

11. A function machine: input → ×4 → output. What is the output when the input is 3/4?

12. A function machine: input → ×0.5 → output. What is the output when the input is 19? A) 8.5 B) 9 C) 9.5 D) 10 E) 10.5

13. A machine trebles a number, then subtracts 6. If the result is 18, what was the original number?

14. A function machine: input → ÷5 → +8 → output. What is the output when the input is −10?

15. A function machine takes a number, adds 1/3 to it, then triples the result. What is the output when the input is 2/3?

Standard / Advanced (15 questions)

1. A function machine doubles a number and adds 6 to give 20. What was the number?

2. A number is tripled, then 9 is subtracted, leaving 15. Find the number.

3. A machine subtracts 5 from a number, then multiplies the result by 4, giving 24. What was the number?

4. A plumber charges a £15 call-out fee plus £25 for each hour worked. A job costs £90 in total. How many hours did the plumber work?

5. A number has 4 added to it, and the result is then doubled to give 30. What was the original number?

6. A function machine applies the rule "add 8, then multiply by 3, then subtract 10" to its input. What is the output when the input is 2?

7. A theme park charges a £6 entry fee plus £2 per ride. Modelled as a function machine (input = number of rides, output = total cost), Freya pays £24 in total. How many rides did she go on? A) 7 B) 8 C) 9 D) 10 E) 11

8. A function machine's rule is "multiply by 2, then add 7, then multiply by 2 again". What is the output when the input is 5?

9. A function machine turns 3 into 11 and 6 into 20 using the rule "×a, then +b". What is the output when the input is 10? A) 28 B) 30 C) 32 D) 34 E) 36

10. A function machine applies the rule "subtract 6, then halve, then add 9" to its input. The output is 16. Find the input.

11. Select ALL that apply: a function machine applies the rule "add 5, then double, then subtract 3" to inputs. Which input→output pairs are correct? A) 2→11 B) 4→16 C) 6→19 D) 8→24 E) 10→27

12. A delivery company charges a £3.50 base fee plus £0.80 per kilometre. Modelled as a function machine (input = distance in km, output = total charge), a delivery costs £11.50. How far was the delivery, in km?

13. A function machine applies the two-step rule "treble the input, then subtract 7" to turn 9 into 20. A second function machine applies the rule "double the input, then add 5" to the same starting number, 9. What is the output of the second machine?

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14. A number is put into a machine that adds 9, and the result is then put into a second machine that multiplies by 2 and subtracts 3. If the original number is 5, what is the final output? A) 23 B) 25 C) 27 D) 29 E) 31

15. A function machine repeats the rule "multiply by 2, then subtract 3" twice in a row (apply the whole machine, then pass its output through the same machine again). What is the final output when the starting input is 4?

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